

Beier Zhu

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Abbreviations

NTU	Nanyang Technological University	MM	ACM International Conference on Multimedia
THU	Tsinghua University	ICCV	IEEE International Conference on Computer Vision
USTC	University of Science and Technology of China	AAAI	AAAI Conference on Artificial Intelligence
CVPR	Conference on Computer Vision and Pattern Recognition	TIP	IEEE Transactions on Image Processing
NeurIPS	Conference on Neural Information Processing Systems	TSG	IEEE Transactions on Smart Grid
ICLR	International Conference on Learning Representations	EE	Electrical Engineering

Research Interests	Robust and fair learning with provable guarantees , including imbalanced learning, group robustness, OOD generalization, and fast diffusion solvers.	
	Multimodal foundation models (VLMs, MLLMs, diffusion models) with a focus on robust adaptation, faithful reasoning, and controllable generation.	
Working Experience	Professor, USTC, China	Mar 2026 – Present
	Research Scientist, NTU, Singapore	Mar 2025 – Feb 2026
	Researcher, SenseTime, Beijing	Jul 2019 – Nov 2020
Education	Ph.D. in Computer Science, NTU, Singapore <i>supervisor: Prof. Hanwang Zhang</i>	Jan 2021 – Jan 2025
	M.S. in EE, THU, Beijing.	Sep 2016 – Jul 2019
	B.E. in EE, THU, Beijing.	Sep 2012 – Jul 2016

Selected Publications Listed in chronological order. *: equal contribution. †: corresponding author.

First-Authored Publications

TPAMI 2026	1. Parallel diffusion solver via residual dirichlet policy optimization. <i>R. Wang*, Z. Li*, B. Zhu*, L. Yuan, H. Zhang, X. Yang, X. Chang, C. Zhang</i>
ICLR 2026	2. Reducing class-wise performance disparity via margin regularization. <i>B. Zhu, K. Zhao, J. Cui, Q. Sun, Y. Zhou, X. Yang, H. Zhang</i>
NeurIPS 2025	3. Adaptive stochastic coefficients for accelerating diffusion sampling. <i>R. Wang*, B. Zhu*, J. Li, L. Yuan, C. Zhang</i>
ICCV 2025	4. Distilling parallel gradients for fast ODE solvers of diffusion models. <i>B. Zhu*, R. Wang*, T. Zhao, H. Zhang, C. Zhang</i>
CVPR 2025 Highlight	5. Project-probe-aggregate: efficient fine-tuning for group robustness. <i>B. Zhu, J. Cui, H. Zhang, C. Zhang</i>
NeurIPS 2024	6. Robust fine-tuning of zero-shot models via variance reduction. <i>B. Zhu, J. Cui, H. Zhang</i>
NeurIPS 2024 Spotlight	7. Enhancing zero-shot vision models by label-free prompt distribution learning and bias correcting. <i>X. Zhu*, B. Zhu*, Y. Tan, S. Wang, Y. Hao, H. Zhang</i>
MM 2024 Oral	8. Selective vision-Language subspace projection for few-shot CLIP. <i>X. Zhu*, B. Zhu*, Y. Tan, S. Wang, Y. Hao, H. Zhang</i>
NeurIPS 2023	9. Generalized logit adjustment: calibrating fine-tuned models by removing label bias in foundation models. <i>B. Zhu, K. Tang, Q. Sun and H. Zhang</i>
ICCV 2023	10. Prompt-aligned gradient for prompt tuning. <i>B. Zhu, Y. Niu, Y. Han, Y. Wu and H. Zhang</i>
AAAI 2023 Oral	11. Debiased fine-tuning for vision-language models by prompt regularization. <i>B. Zhu, Y. Niu, S. Lee, M. Hur, H. Zhang</i>
AAAI 2022 Oral	12. Cross-domain empirical risk minimization for unbiased long-tailed classification. <i>B. Zhu, Y. Niu, X. Hua and H. Zhang</i>
US patent 2022	13. Polygonal region detection. <i>Beier Zhu, Rui Zhang</i>
TIP 2021	14. Structure-coherent deep feature learning for robust face alignment. <i>C. Lin*, B. Zhu*, Q. Wang, R. Liao, C. Qian, J. Lu and J. Zhou</i>

Corresponding-Authored Publications

- ICML 2026** 15. Omni-perception policy optimization for multimodal emotion reasoning.
Z. Han, †B. Zhu, W. Tong, P. Shao, P. Song, X. Wang, J. Chen, L. Lu, X. Yang
- CVPR 2026** 16. Adapting point cloud analysis via multimodal Bayesian distribution learning.
X. Zhu, L. Yi, S. Wang, W. Zhu, Y. Wu, †B. Zhu, H. Zhang
- TIP 2026** 17. Hybrid granularity distribution estimation for few-shot learning: statistics transfer from categories and instances.
S. Wang, T. Qi, X. Zhu, Y. Hao, †B. Zhu, H. Zhang, M. Wang
- ICLR 2026** 18. Real-time motion-controllable autoregressive video diffusion.
K. Zhao, J. Shi, †B. Zhu, J. Zhou, X. Shen, Y. Zhou, Q. Sun, H. Zhang
- ICLR 2026** 19. Look carefully: adaptive visual reinforcements in multimodal large language models for hallucination mitigation.
X. Zhu, K. Zhao, L. Yi, S. Wang, Z. Wang, †B. Zhu, H. Zhang, X. He
- AAAI 2026** 20. DEPO: dual-efficiency preference optimization for LLM agents.
S. Chen, M. Zhao, L. Xu, Y. Zhao, †B. Zhu, H. Zhang, S. Zhao, C. Lu
- MM 2025** 21. Benchmarking and bridging emotion conflicts for multimodal emotion reasoning.
Z. Han, †B. Zhu, Y. Xu, P. Song, X. Yang
- Oral**
- ICCV 2025** 22. Unsupervised visual chain-of-thought reasoning via preference optimization.
K. Zhao, †B. Zhu, Q. Sun, H. Zhang
- TSG 2019** 23. Fault location for radial distribution network via topology and reclosure-generating traveling waves.
S. Shi, †B. Zhu, A. Lei and X. Dong
- TSG 2018** 24. Fault classification for transmission lines based on group sparse representation.
S. Shi, †B. Zhu, S. Mirsaeidi and X. Dong

Others

- ACL 2026 Findings** 25. AMA: adaptive memory via multi-agent collaboration.
W. Huang, Z. Wang, H. Lin, S. Wang, B. Xu, Q. Li, B. Zhu, L. Yang, C. Qin
- ACL 2026** 26. Mitigating hallucinations in large vision-language models without performance degradation.
X. Zhu, J. Fang, S. Wang, B. Zhu, Z. Wang, Y. Yang, X. He
- ICML 2026** 27. Robustifying vision-language models via test-time prompt adaptation.
X. Zhu, H. Wu, S. Wang, B. Zhu, and J. Ge, J. Zhang, L. Chen
- CVPR 2026** 28. Principled steering via null-space projection for jailbreak defense in vision-language models
X. Zhu, B. Zhu, S. Wang, J. Fang, K. Zhao, H. Zhang, X. He
- CVPR 2026** 29. Few-step diffusion sampling through instance-aware discretizations
L. Yuan, R. Wang, T. Zhao, D. Fu, M. Lei, B. Zhu, C. Zhang
- CVPR 2026** 30. ACE-merging: data-free model merging with adaptive covariance estimation
B. Xu, H. Wu, H. Lin, W. Huang, B. Zhu, Y. Shu, C. Qin
- ICLR 2026** 31. CoDi: subject-consistent and pose-diverse text-to-image generation.
Z. Gao, B. Zhu, L. Yao, J. Yang, Y. Tai
- ICLR 2026** 32. GuardAlign: robust safety alignment in multimodal large language models.
X. Zhu, B. Zhu, J. Fang, S. Wang, Y. Zhang, X. Wang, X. He
- ICLR 2026** 33. Free lunch for stabilizing rectified flow inversion.
C. Wang, B. Zhu, C. Zhang
- ICLR 2026** 34. Streaming drag-oriented interactive video manipulation: drag anything, anytime!
J. Zhou, Y. Zhou, K. Zhao, Q. Xu, B. Zhu, R. Hong, H. Zhang
- AAAI 2026** 35. Hierarchical semantic alignment for image clustering.
X. Zhu, B. Zhu, Y. Li, J. Fang, S. Wang, K. Zhao, H. Zhang
- Preprint 2025** 36. Generative distribution distillation.
J. Cui, B. Zhu, Q. Xu, X. Xu, P. Chen, X. Qi, B. Yu, H. Zhang, R. Hong
- Preprint 2025** 37. Generalized kullback-leibler divergence loss.
J. Cui, B. Zhu, Q. Xu, Z. Tian, X. Qi, B. Yu, H. Zhang, R. Hong
- NeurIPS 2025 Spotlight** 38. Enhancing CLIP robustness via cross-modality alignment.
X. Zhu, B. Zhu, S. Wang, K. Zhao, H. Zhang
- ICCV 2025** 39. Dynamic multimodal prototype learning in vision-language models.
X. Zhu, S. Wang, B. Zhu, M. Li, Y. Li, J. Fang, Z. Wang, D. Wang, H. Zhang
- CVPR 2025** 40. Devils in middle layers of large vision-language models: interpreting, detecting and mitigating object hallucinations via attention lens.

	Z. Jiang, J. Chen, B. Zhu , T. Luo, Y. Shen, X. Yang	
CVPR 2025	41. Stylestudio: text-driven style transfer with selective control of style elements.	
	M. Lei, X. Song, B. Zhu , H. Wang, C. Zhang	
CVPR 2024	42. Classes are not equal: an empirical study on image recognition fairness.	
	J. Cui, B. Zhu , X. Wen, X. Qi, B. Yu, H. Zhang	
AAAI 2023	43. Leveraging modality-specific representations for audio-visual speech recognition via reinforcement learning.	
Oral	C. Chen, Y. Hu, Q. Zhang, H. Zou, B. Zhu , E. Chng	
Mentoring and Supervision	Xingyu Zhu , Ph.D at USTC <i>Supervisor for papers accepted at MM 2024, NeurIPS 2024/2025, ICCV 2025, AAAI 2026, and ICLR 2026.</i> Kesen Zhao , Ph.D. at NTU <i>Supervisor for paper accepted at ICCV 2025 and ICLR 2026.</i> Ruoyu Wang , Ph.D. at Westlake University <i>Supervisor for papers accepted at ICCV 2025, NeurIPS 2025 and TPAMI 2026.</i> Zhiyuan Han , Ph.D. at USTC <i>Supervisor for paper accepted at MM 2025 and ICML 2026.</i> Sirui Chen , Ph.D. at Tongji University <i>Supervisor for paper accepted at AAAI 2026.</i> Zhanxin Gao , Ph.D. at Nanjing University <i>Supervisor for paper accepted at ICLR 2026.</i>	
Service	<i>Journal Reviewer:</i> TPAMI, TIP, TMM, TCSVT, TCE, IJCV, Information Fusion <i>Conference Reviewer:</i> CVPR, ICCV, AAAI, MM, UAI, AISTATS, NeurIPS (Top Reviewer 2025), ICLR, ICML, ACL.	
Awards	<i>Honorable Mention</i> , Nanyang Speech Forum, Singapore. 2023 <i>AISG PhD Fellowship</i> , National Research Foundation, Singapore. 2021 <i>First Price Scholarship</i> , THU, China. 2018 <i>Scholarship</i> , China Scholarship Council, China. 2014	
Internship	Research Intern, SenseTime, Beijing Mar 2018 – Oct 2018 Research Intern, Tencent YouTu Lab, Shanghai Jul 2018 – Sep 2018 Research Intern, MeiTuan, Beijing Aug 2017 – Dec 2017	

[CV compiled on 2026-05-13]